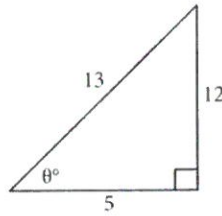


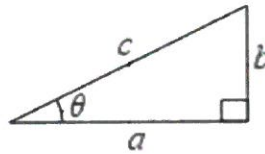
Question 4. What is the correct ratio to find $\tan \theta$

- A. $13/12$
- B. $5/12$
- C. $5/13$
- ☒ D. $12/5$



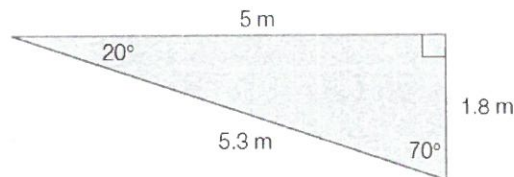
Question 5. Which statement is true for the angle θ in the diagram?

- ☒ A. $b = \text{opposite}$
- B. $b = \text{adjacent}$
- C. $c = \text{adjacent}$
- D. $a = \text{opposite}$



Question 6. What is the length of the adjacent side when considering the 20° angle?

- A. 20
- ☒ B. 5
- C. 1.8
- D. 5.3



Short answer questions

39 marks

Show FULL working out to get FULL marks

Question 7. Complete the following using words from the list below.

[4 marks]

adjacent angle of depression angle of elevation cosine
hypotenuse opposite sine tangent

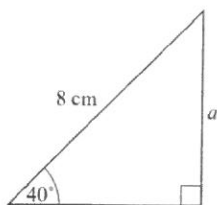
The sine ratio is found by dividing the opposite side by the hypotenuse.

The tangent ratio is found by dividing the opposite side by the hypotenuse side.

When you look up at an aeroplane the angle between the horizontal and your line of sight is called the angle of elevation

Question 8. Calculate the lengths of the sides marked with letters in each of the following triangles, giving your answers to 3 decimal places. [6 marks]

(a)

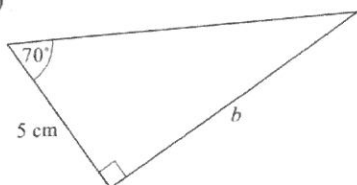


$$\sin 40 = \frac{a}{8} \quad \checkmark$$

$$8 \times \sin 40 = a$$

$$5.142 \text{ cm} = a \quad \checkmark$$

(b)

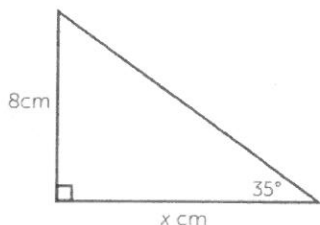


$$\tan 70 = \frac{b}{5} \quad \checkmark$$

$$5 \times \tan 70 = b$$

$$13.737 \text{ cm} = b \quad \checkmark$$

(c)



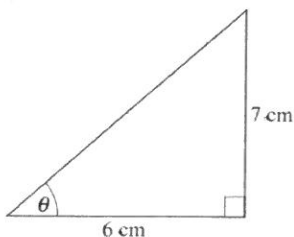
$$\tan 35 = \frac{8}{x} \quad \checkmark$$

$$x = \frac{8}{\tan 35}$$

$$x = 11.425 \text{ cm} \quad \checkmark$$

Question 9. Calculate the size of each of the angles marked θ in the following triangles, giving your answers to the nearest degree. [4 marks]

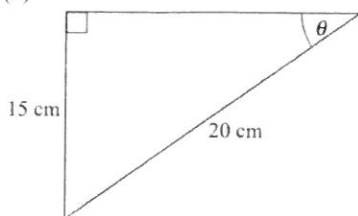
(a)



$$\theta = \tan^{-1}\left(\frac{7}{6}\right) \quad \checkmark$$

$$= 49^\circ \quad \checkmark$$

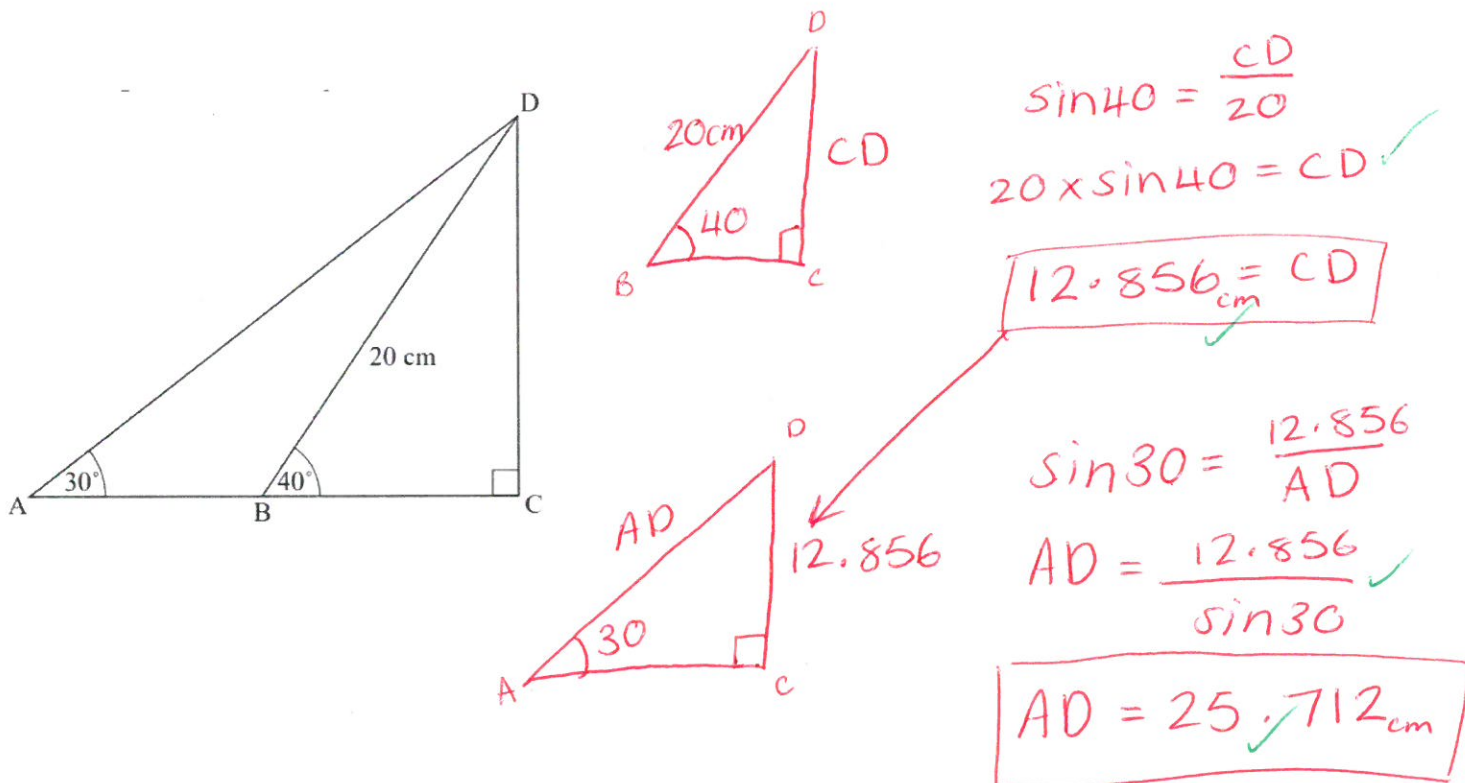
(b)



$$\theta = \sin^{-1}\left(\frac{15}{20}\right) \quad \checkmark$$

$$= 49^\circ \quad \checkmark$$

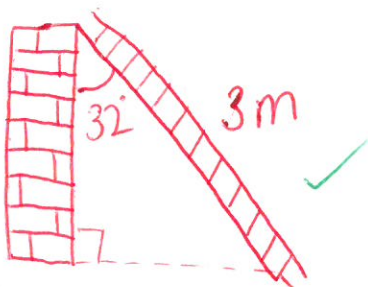
Question 10. For the triangle shown, calculate the length of **CD** and **AD**. Give your answers to 3 decimal places. **[4 marks]**



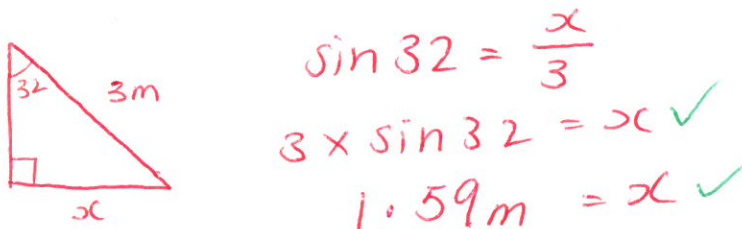
Question 11. A ladder of length 3 m makes an angle of 32° with the wall.

[5 marks]

a) Draw a diagram and label the given values.



b) How far is the base of the ladder from the wall?



c) What angle does the ladder make with the ground?

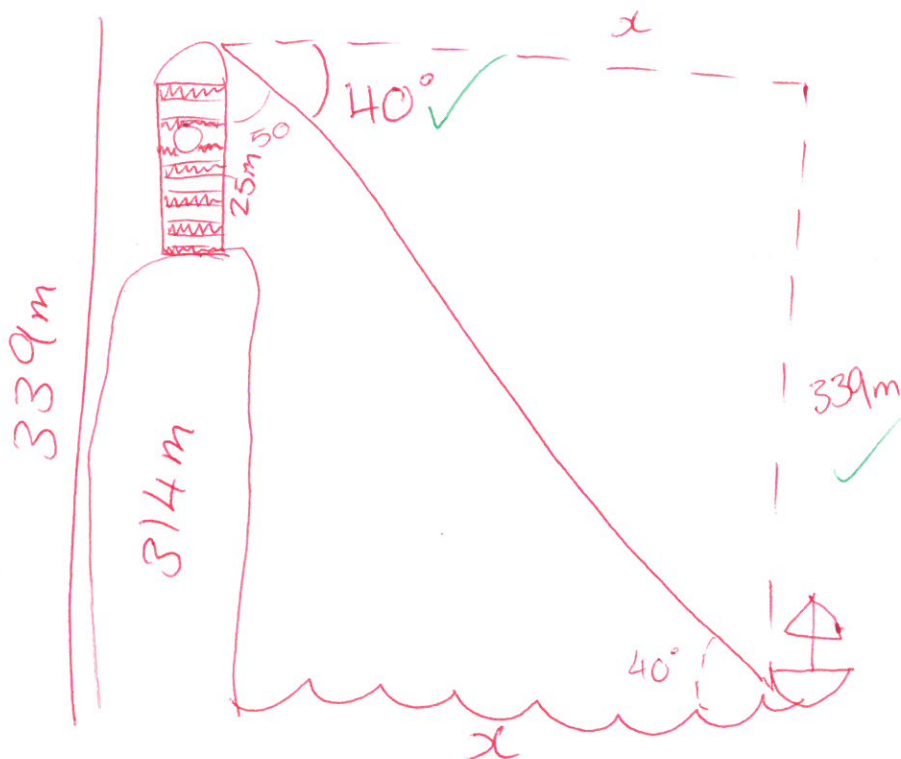
Handwritten calculations for Question 11c:

$$180 - (90 + 32) = \theta$$

$$58^\circ = \theta$$

Question 12. From the top of a 25 m lighthouse, on a 314 m tall cliff, the angle of depression to a sailing boat out in the ocean is 40° . How far is the sailing boat from the base of the cliff to the nearest metre?

[4 marks]



$$\tan 40 = \frac{339}{x}$$

$$x = \frac{339}{\tan 40}$$

$$x = 404 \text{ m}$$

OR

$$\tan 50 = \frac{x}{339}$$

$$339 \times \tan 50 = x$$

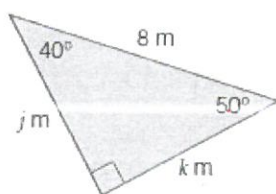
$$404 \text{ m} = x$$

Question 13. The following answers were given by a student on a trigonometry test.

[6 marks]

(i) Find the value of k .

(ii) Find the value of x .

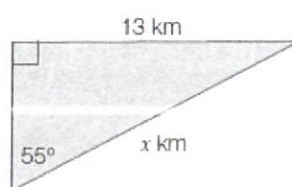


$$\cos \theta = \frac{A}{H}$$

$$\cos 40^\circ = \frac{k}{8}$$

$$8 \times \cos 40^\circ = k$$

$$k = 6.13$$



$$\sin \theta = \frac{O}{H}$$

$$\sin 55^\circ = \frac{13}{x}$$

$$13 \times \sin 55^\circ = x$$

$$x = 10.65 \text{ km}$$

a) Explain the mistake the student has made in each question

(i) used wrong angle OR used wrong ratio; (should be $\sin 40$ or $\cos 50$)

(ii) multiplied the ratio & opposite instead of dividing.

b) Show the correct calculations and answers for each question.

$$(i) \sin 40 = \frac{k}{8}$$

$$k = 5.14 \text{ m}$$

$$\cos 50 = \frac{k}{8}$$

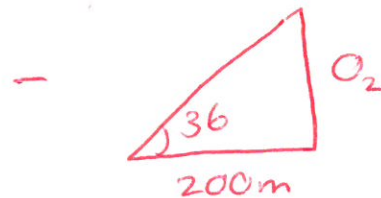
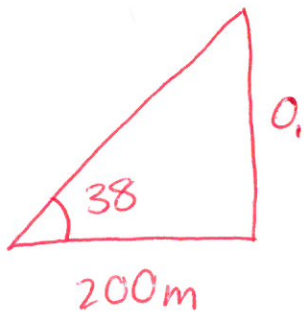
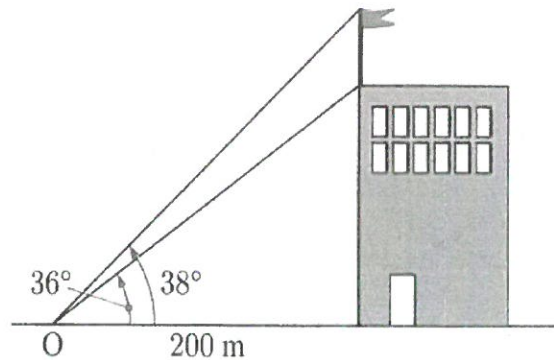
$$k = 5.14 \text{ m}$$

$$(ii) \sin 55 = \frac{13}{x}$$

$$x = \frac{13}{\sin 55}$$

$$x = 15.87 \text{ km}$$

Question 14. From an observer at O who is 200m from a building, the angles of elevation to the bottom and top of a flagpole are 36° and 38° respectively. Find the height of the flagpole to the nearest hundredth. [6 marks]



$$\tan 38 = \frac{O_1}{200} \quad \checkmark$$

$$200 \times \tan 38 = O_1$$

$$156.26 = O_1 \quad \checkmark$$

$$\tan 36 = \frac{O_2}{200} \quad \checkmark$$

$$200 \times \tan 36 = O_2$$

$$145.31 = O_2 \quad \checkmark$$

$$O_1 - O_2 = \text{Flagpole height} \quad \checkmark$$

$$156.26 - 145.31 = \text{Flagpole height} \quad \checkmark$$

$$10.95 \text{ m} = \text{Flagpole height} \quad \checkmark$$

END OF TEST ☺